

SOLUTION

Solution Q1

- Local variable inside function overrides global variable
- This follows static scope rule (search starts from local scope)

✅ Answer: Local variable is used

👉 Based on scope rules from Week 5 (names & scope)

Solution Q2

(a) Output:

5

(b) Explanation:

- Function parameter a is a local copy
- Changes inside fun() do not affect original a
- This is pass-by-value binding

👉 Binding occurs at function call (runtime)

Solution Q3

(a) Evaluation:

$$result = 2 + (3 \times 4) = 2 + 12 = 14$$

(b) Modify:

int result = (a + b) * c;

(c) Explanation:

- * has higher precedence than +
- So multiplication happens first unless parentheses override

👉 From expression rules (Week 9)

Solution Q4

(a) Evaluation:

$$(\lambda(x) \ x * x)(3) = 3 * 3 = 9$$

(b) Explanation:

- Lambda defines a function without name
- Value is computed directly without variables/state
- Unlike imperative programming, no assignment is needed